ConvertAudio2-DDP51.ps1 (v1.0.5)

> If your MKV movie has two English audio tracks (e.g., TrueHD 7.1 and DTS-HD 5.1),
> ConvertAudio2-DDP51 keeps only the best one (TrueHD 7.1 → downmix).
> The other audio track is removed.

All stereo (2.0) tracks are removed. Stereo output is handled separately by ConvertAudio2-Stereo.ps1

- **PreTrack-QRS** — selects the best source per language using priority ranking and a clear tie-break order (rank → index) before any encoding decision is made.
- **Safe Peak Normalizer (SPN)** — scans first, applies loudnorm only when the source truly needs it, preserving dynamics on clean tracks.
- **3-Phase Pipeline (A/B/C)** — sequential rule matching, parallel peak scanning, and sequential result merging for reliable output.
- **Malformed-layout correction** — fixes invalid 7-channel layouts before processing.
- **ITU-R BS.775 pan matrix** — accurate 7.1 → 5.1 downmixing.
- **Per-language output** — one DD+ 5.1 track per language group in the source file.

PowerShell 7.6 - [Click here to Download PS 7.6 (Windows x64 .msi)]()
FFmpeg 8.1 - https://ffmpeg.org/download.html

Quick Start (Windows 11)

1. Place `ConvertAudio2-DDP51.ps1`, `ffmpeg.exe`, `ffprobe.exe` in the same folder.
2. Run the script:

> pwsh -ExecutionPolicy Bypass -File .\ConvertAudio2-DDP51.ps1 ".\YourMovie.mkv"

The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

macOS / Linux

1. Place the script, `ffmpeg`, and `ffprobe` in the same folder.
2. Make the binaries executable (first time only):

```
chmod +x ffmpeg ffprobe
```

3. Run:

```
pwsh -File ./ConvertAudio2-DDP51.ps1 "./YourMovie.mkv"
```

Output filename: The program saves a new file in the same folder, adding a codec suffix like **_ddp5only.mkv**. Your original file stays untouched.

1. To Adjust Bitrate (line 54)

```
You can safely adjust these numbers, within '....k'
Common EAC3 5.1 bitrates: 384, 448, 512, 640, 768, 896
Higher pipeline bitrates: 1024, 1152, 1280, 1408, 1536

$BitRateConfig = @{
    Downmix51   = '1152k'   << Change your Bitrate
    ReEncode51  = '768k'    << Change your Bitrate
    EncodeDTSHD = '1024k'   << Change your Bitrate
```

Remember to save the file after making changes.

Bitrates are set in the `$BitRateConfig` table near the top of the script.
All three values can be adjusted safely, changes apply automatically throughout the script.

Key	Default	Used for
Downmix51	1152k	All 7.1 sources downmixed to 5.1
ReEncode51	768k	5.1 sources re-encoded to DD+
EncodeDTSHD	1024k	DTS-HD MA and HRA sources (5.1 and 7.1)

2. Supported Codec Families

Family	Formats
EAC3 / Atmos	EAC3, EAC3-JOC (Atmos)
TrueHD	TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos
DTS	DTS Core, DTS-HD MA, DTS-HD HRA
AAC	AAC 5.1, AAC 7.1
PCM	pcm_s16le, pcm_s24le, pcm_f32le, pcm_f32be
FLAC	All FLAC multichannel layouts

All stereo (2.0) tracks from any codec family are removed.

3. What Happens to Each Track

The script evaluates every audio track and assigns one of four actions.

Action	Color	Meaning
Passthrough	Green	Track is already DD+ EAC3 5.1 or Atmos — copied without re-encoding
Downmix	Yellow	7.1 source — downmixed to 5.1 and encoded to DD+
Encode	Blue	5.1 source in another codec — re-encoded to DD+
Removed	Red	2.0 stereo, commentary, or unsupported layout — excluded from output

- EAC3 Atmos (JOC) tracks at any channel count above 2.0 are passed through at the highest priority (100).
- EAC3 5.1 tracks are passed through as-is.
- TrueHD 5.1 tracks are copied losslessly (passthrough).
- No -AC-3 core extraction is performed.
- All other multichannel codecs are encoded to DD+ EAC3.

4. Audio Rule Priority Table

Higher priority = selected first.
The engine picks the highest-priority rule that matches the track's codec, channel count, and profile.

Priority	Format	Action	Bitrate
100	EAC3 Atmos (JOC) any ch > 2	Passthrough	—
99	EAC3 7.1	Downmix	Downmix51
98	EAC3 5.1	Passthrough	—
97	EAC3 2.0	Removed	—
92	AAC 7.1	Downmix	Downmix51
91	AAC 5.1	Encode	ReEncode51
90	AAC 2.0	Removed	—
81	TrueHD 5.1	Passthrough	—
80	TrueHD 7.1	Downmix	Downmix51
73	DTS-HD MA/HRA (any ch > 2)	Encode	EncodeDTSHD
72	DTS Core (any ch > 2)	Encode	ReEncode51
71	DTS-HD 2.0	Removed	—
70	DTS 2.0	Removed	—
62	PCM/FLAC 7.1	Downmix	Downmix51
61	PCM/FLAC 5.1	Encode	ReEncode51
60	PCM/FLAC 2.0	Removed	—

Downmix51 -> Default value = 1152k
ReEncode51 -> Default value = 768k
EncodeDTSHD -> Default value = 1024k

5. PreTrack-QRS (Per Language Winner Selection)

When a file has more than one multichannel track of the same language, PreTrack-QRS picks the best one and removes the rest before scanning begins.

PreTrack-QRS runs **after Phase A** (rule matching, malformed-layout correction, commentary removal) and **before Phase B** (SPN peak scanning).
Its purpose is to ensure that only the *best* multichannel track for each language survives into the SPN and encoding stages.
This prevents duplicate outputs and secures consistent behavior across all runs.

How it works:

1. All tracks already removed (2.0, commentary, malformed) are skipped.
2. Remaining tracks are grouped by language.
3. Within each language group, the track with the highest Priority wins.
4. If two tracks share the same Priority, the one with the lower stream index wins.
5. All losing tracks in that language group are marked Removed before Phase B (SPN) runs.

This ensures the SPN does not waste time scanning tracks that will not be used, and that the output contains exactly one DD+ 5.1 track per language.

Why this matters:

- Prevents duplicate DD+ 5.1 tracks for the same language.
- Ensures the highest quality source (TrueHD > DTS-HD >> AAC) is always selected.
- Reduces SPN workload by scanning only the surviving tracks.
- Produces stable, reproducible results regardless of track order or codec mix.

What QRS does not do:

- It does not override malformed-layout guards or commentary removal.
- It does not change codec actions (Passthrough, Downmix, Encode).
- It does not modify rule priorities.

QRS only determines **which** track survives within each language group. The rest of the pipeline determines **how** that track is processed.

```
### 3-phase A/B/C architecture
Input MKV >> Probe
     > Phase A (rules + malformed + commentary)
     > PreTrack-QRS
     > Phase B (SPN peak scan)
     > Phase C (merge)
     > FFmpeg command builder
       >> Output MKV
```

6. FFmpeg Global Options

Flag	Purpose
-y	Overwrites the output file without asking.
-loglevel error	Hides all FFmpeg output except actual errors.
-analyzeduration 200M	Deep probe for TrueHD/Atmos/AAC 7.1 layouts.
-probesize 200M	Prevents mis-detected channel layouts.
-drc_scale 0	Disables Dolby DRC for consistent loudness.
-max_muxing_queue_size	**(14000)** Prevents muxer stalls on long files.
-map 0:v? -c:v copy	Always copy video losslessly (no re-encode).
-map_chapters 0	**(-map_metadata 0)** Preserve metadata and chapters.
-dialnorm -31	Dolby loudness signaling. Disables receiver-side volume adjustment.
-cutoff 20000	Sets audio bandwidth to 20 kHz. Applies to downmix and re-encode paths only. Not passthrough.
-ac 6	Forces 6-channel output on the 5.1 re-encode path. Ensures correct channel count even if the source reports a non-standard layout.
`-disposition:a:$i`	Sets the first kept audio track as default. All others are set to none. Ensures players select the correct track automatically.

> **Subtitles and attachments** are always carried through unchanged. The script maps `-map 0:s?` (all subtitles) and `-map 0:t?` (all attachments — fonts, cover art) with stream copy. There is no toggle to disable this.

7. ScanThrottle + ThreadCount

Sets how many CPU threads the encoder uses (line 35).

```
$ThreadCount = 8    # User-adjustable (4-16).
```

Limits how many peak-scan tasks run at the same time (line 37).

```
$ScanThrottle = [Math]::Max(1, [int]([Environment]::ProcessorCount /
$ThreadCount))
# Recommended overrides: 1, 2, 3, or 4
```

What ScanThrottle does:

- Limits parallel SPN scans to avoid CPU overload.
- Keeps the system responsive during peak detection.
- Does not affect encoding speed.
- SSD/NVMe: safe to override up to 4. Spinning disk: keep at 1.

8. Safe Peak Normalizer (SPN)

SPN is a pre-filter safety system that checks each audio track for unsafe peaks before any downmix or encode occurs.

What SPN does:

- Scans each non-Removed track using `volumedetect`.
- Compares the detected peak against **−0.5 dBFS**.
- Marks the track with **NeedsNormalization** when the peak is too high.
- Inserts a light `loudnorm` pass only when required.
- Clean sources are left untouched.
- Dirty sources are corrected safely without compression artifacts.
- Passthrough tracks are skipped entirely.

SPN Behavior

Condition	Behavior
Peak **greater than** −0.5 dBFS	Apply `loudnorm` before encode
Peak **stays at or under** −0.5 dBFS	No normalization
Passthrough track	SPN skipped
QRS-removed track	SPN skipped

SPN Thresholds (line 41)

These are the values Safe Peak Normalizer uses to decide when to step in.

```
$PeakThresholdDB = -0.5 # SPN: loudnorm triggers when source peak exceeds this (dBFS)
```

- Valid range: **−1.5 dBFS to −0.3 dBFS** (recommended: −0.5 dBFS)
- Lower values (e.g. −1.5) normalize more tracks — more conservative.
- Higher values (e.g. −0.3) normalize fewer tracks — only the hottest sources.

Setting	Value
loudnorm target	−23 LUFS
Loudness range target	LRA=11 LU
True peak ceiling	−1.5 dBTP

9. 3-Phase Pipeline

The engine processes tracks in three sequential phases.

Phase A — Sequential rule matching
Every audio stream is probed and matched against the rule table. Malformed-layout guards and commentary detection run here. 2.0 tracks are marked Removed. Surviving tracks get a pending action (Passthrough, Downmix, Encode).

DTS-HD MA and HRA tracks are always re-encoded to DD+ using `EncodeDTSHD` (1024k), regardless of whether the source is 5.1 or 7.1. This is intentional, DTS-HD is a lossless container that requires transcoding to produce a compatible EAC3 output.

PreTrack-QRS runs between Phase A and Phase B. Same-language losers are removed before scanning begins.

Phase B — Parallel SPN peak scans
All non-Removed, non-Passthrough tracks are scanned in parallel for peak levels. Results are collected during the parallel scan; the track list is not modified in this phase.

Phase C — Sequential merge
Peak results are merged back into the track list. NeedsNormalization is resolved from the scan data. The final track list is ready for FFmpeg.

10. 7.1 → 5.1 Downmix Matrix (ITU-R BS.775)

When a 7.1 source is downmixed to 5.1, the side channels (SL/SR) are folded into the rear channels (BL/BR) at −3 dB using the ITU-R BS.775 pan matrix.

```
aformat=channel_layouts=7.1,
pan=5.1|FL=FL|FR=FR|FC=FC|LFE=LFE|BL=BL+0.707*SL|BR=BR+0.707*SR,
alimiter=limit=0.948:attack=5:release=50:level=disabled:latency=1
```

- **0.707** = −3 dB fold for side channels (ITU-R standard).
- **aformat** ensures a consistent 7.1 layout before the pan matrix runs.
- **alimiter** catches post-pan true-peak overshoots (attack 5 ms, release 50 ms).
- **latency=1** allows the limiter a 1-sample look-ahead window to catch post-downmix peaks without adding audible delay.

The limiter is always applied last in the audio chain.

> **SPN threshold vs. limiter ceiling — these are two separate systems.**
> SPN (`$PeakThresholdDB = -0.5`) decides whether `loudnorm` is applied before encoding.
> The alimiter (`limit=0.948` , ≈ −0.47 dBFS) runs after the pan matrix and catches any peaks introduced by channel folding. They operate independently.

11. Malformed-Layout Guards

Some sources report invalid 7-channel layouts. The engine corrects these before rule matching and QRS run.

Codec	7ch detected → corrected to	Result
AAC	7ch → 6ch	Treated as 5.1
EAC3	7ch → 6ch	Treated as 5.1
TrueHD	7ch → 8ch	Treated as 7.1
PCM/FLAC	7ch → 8ch	Treated as 7.1
AC3	7ch	Removed (AC3 7ch has no valid layout)

AAC 7ch is extremely rare in practice and almost always indicates a malformed or mislabeled stream. The engine silently corrects it to 6ch before rule matching. No user action is required.

12. Commentary Detection

Tracks are marked as Commentary and removed when their title metadata matches the commentary pattern and the track has exactly 2 channels.

Keywords detected: `director, producer, writer, cast, behind, bonus, alt, interview, commentary`

Commentary detection applies only to 2-channel tracks. Multichannel tracks with these keywords in the title are not affected.

13. Safety Audit (Bitrate Clamps)

After rule matching, the engine checks that each track's assigned bitrate is appropriate for its channel count. If a mismatch is detected (e.g., a 6ch track was assigned the 7.1 downmix bitrate), the bitrate is clamped to the correct value and a warning is printed.

Condition	Clamp applied
6ch track assigned Downmix51 bitrate	Clamped to ReEncode51
7ch track assigned ReEncode51 bitrate	Clamped to Downmix51
8ch track assigned ReEncode51 bitrate	Clamped to Downmix51

DTS-HD MA/HRA at **EncodeDTSHD** bitrate is exempt from the 6ch clamp. It is a legitimate lossless 5.1 encode.

14. Fallback Logic

If no rule in the table matches a track's codec and channel count, the engine applies a safe fallback.

Channel count	Fallback action	Bitrate
2 channels or fewer	Removed	—
5.1 (6 channels)	Encode	ReEncode51
more than 6 channels (7ch+)	Downmix	Downmix51

This guarantees a valid output even on unknown or future codec variants.

15. Audio Processing Summary

The script generates a detailed table with the following columns:

Column	Description
Idx	Internal track index (0, 1, 2 etc).
Codec	Source codec (TrueHD, EAC3, DTS, AAC, etc.).
Ch	Corrected channel count (after malformed-layout guard).
Action	Removed / Encode / Downmix / Passthrough with bitrate.
Output	Final codec + bitrate + metadata tag.
Pri	Rule priority (0–100). Removed tracks show —.
Rule	The rule that matched (or fallback).

Color by action:
Red = Removed
Yellow = Downmix
Blue = Encode
Green = Passthrough

16. Re-Enabling 2.0 Encoding (Optional)

All 2.0 rules are present in the engine but set to `Action = "Removed"` by default. This is intentional. Stereo output is delegated to **ConvertAudio2-Stereo**. If you want this script to encode 2.0 tracks instead of removing them, you can re-enable any rule by changing three fields.

Example — re-enabling AAC 2.0 at 256k:

```
# Find this block in $Rules_AAC (Rule3):
Action              = "Removed"
Bitrate             = $null
Rule                = "AAC_2.0_Removed"

# Change it to:
Action              = "Encode"
Bitrate             = "256k"
Rule                = "AAC_2.0_Encode_256k"
```

Only those three fields need to change. Nothing else in the engine requires modification.

Apply the same pattern to re-enable any other 2.0 rule:

Rule block	Location	Default	Suggested bitrate
AAC 2.0	Rule3 in `$Rules_AAC`	Removed	256k
EAC3 2.0	Rule4 in `$Rules_EAC3_Atmos`	Removed	384k
DTS-HD 2.0	Rule3 in `$Rules_DTS`	Removed	384k
DTS 2.0	Rule4 in `$Rules_DTS`	Removed	256k
PCM/FLAC 2.0	Rule3 in `$Rules_PCMFLAC`	Removed	384k

17. Non-Critical FFmpeg Warnings (Safe to Ignore)

FFmpeg may print warnings during probing or decoding.
These do not affect audio quality or sync.

Warning	Meaning	Impact
quant_step_size larger than huff_lsbs	TrueHD decoder diagnostic	Safe
Codec AVOption drc_scale … has not been used	DRC disabled intentionally	Safe
Subtitle: hdmv_pgs_subtitle unspecified size	PGS metadata missing	Safe
Assuming an incorrectly encoded 7.1 channel layout	AAC 7.1 variant	Safe (layout is corrected)
Could not find codec parameters for stream (pgssub)	PGS subtitle track has incomplete metadata	Safe
Consider increasing analyzeduration/probesize	Generic suggestion caused by incomplete PGS metadata	Safe
unspecified size (PGS)	Subtitle bitmap dimensions not declared	Safe

18. Audio-Script Rules That Must Not Change

a. Malformed-layout guards

AAC 7ch → 6ch
EAC3 7ch → 6ch
TrueHD 7ch → 8ch
PCM/FLAC 7ch → 8ch
AC3 7ch → Removed

b. SPN runs before encoding

Peak scanning completes before any downmix or encode begins.

c. PreTrack-QRS runs after Phase A, before Phase B

All rule matches and malformed-layout guards must be settled before QRS removes language-group losers.

d. Tie-break sequence

Priority (descending) → RealIndex (ascending, lower index wins).

e. Pan-matrix operator

The pan matrix uses the `=` operator to avoid FFmpeg auto-normalization.

f. Limiter position

The alimiter is applied last in the audio chain, after the pan matrix.

g. Fallback rules must remain intact

Guarantees safe behavior on unknown codecs.

h. Commentary detection

Commentary detection applies only to 2-channel tracks.

i. Safety clamps must remain active

Bitrate clamps are the last line of defense against rule mismatches.

Disclaimer

- Always test on a few files before batch-processing your library.
- Keep backups of original media.
- Results may vary depending on source quality and hardware.